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S 640- FENCE, GATES AND GUARD RAIL WORKS

1. SCOPE OF WORK

- 1.1 The work specified in this section shall cover the furnishing of all materials, labor, transportation, equipment and services necessary for the installation of fences, gates, space, and guard rails as shown on the approved Bidder's drawings or as directed by the Owner.
- 1.2 The work performed shall be in accordance with the Specifications. The materials and workmanship for fence and gates shall conform to the following or approved equal standards:
- ASTM (American Society for Testing and Materials)
 - "Industrial Steel Specifications for Fence Posts, Gates and Accessories".
 - JIS (Japanese Industrial Standards)

2. FENCE TYPE

Fencing shall consist of hot dip galvanized steel with a panel height and width of 240 cm and an overall height of 280 cm from the bottom of the panel to the top of barbed wire. The fence shall have three strands of barbed wire mounted on 45 degree extension arms. The upper strand shall be approximately 35 cm out from the fence and 35 cm above the top of the panel. Posts shall be set in concrete.

3. MATERIALS

3.1 Fence Fabric (BRC)

- 3.1.1 Fence fabric shall be hot dip galvanized, 7.00 mm diameter steel wire in both the horizontal and vertical directions, and have a frame of 240 cm x 240 cm nominal width and height. Galvanizing shall be a minimum thickness of 610 gr/m² (BS 729; BS 3083; 1988; ASTM A527; G90)

- 3.1.2 The fence panel shall consist of 25 wires in the vertical direction and 11 wires in the horizontal direction. Fabric steel wires shall have a minimum Φ 3 mm and minimum tensile strength of 4,100 kg/cm².

3.2 Posts, Rails, Braces, Gates and Appurtenances

3.2.1 General

- 1) Square Tube: Hot dip galvanized square tube conforming to ASTM A120-79
- 2) Fittings and Accessories: Bolt & nut, U clip and post. All shall be hot dip galvanized.
- 3) Galvanizing: hot dip galvanized pipe (inside and outside) and H-sections (all surfaces) and other ferrous parts, with a weight of zinc coating not less than 610 gr/m² of thickness.

3.2.2 Sizes of material for posts, rails, braces and gate frames shall conform to the following dimensions or as shown on the approved Bidder's drawings:

- 1) End, Corner and Pull Posts: Square tube 75 x 75 mm and 3 mm thickness.
- 2) Intermediate Posts: Square tube 75 x 75 mm and 3 mm thickness.
- 3) Gate Posts: Square 100 x 100 mm for 2 to 4 meter nominal width of gate, single or double swing, or sliding gates.
- 4) Exterior Frames for Gates: Hot dip galvanized, rectangular tube 100 mm x 50 mm and 3.5 mm of thickness.

3.3 Attachments

- 3.3.1 To End, Corner, Intermediate and Gate Posts: Shall be as recommended by the fence manufacturer or as approved by the Owner.

- 3.3.2 To Intermediate Posts: Fasten fabric with No. 6 gauge galvanized steel wire clips spaced not more than 375 mm center to center.

3.4 Barbed Wire and Attachments

- 3.4.1 Three lines of two strand 2.5 mm diameter steel wire with 2.11 mm diameter four points barbs spaced approximately 125 mm on center.

All wire shall be hot dip galvanized, with the weight of zinc coating per square meter of surface area not less than 0.24 kg/m² for 2.5 mm diameter wire and not less than 0.20 kg/m² for 2.11 mm diameter wire.

- 3.4.2 Barbed wire shall be fastened to extension arms with heavy galvanized steel pins, and to post extensions with galvanized steel bands.
Refer to the Code number ASTM A123, A153 or JIS H 8641.

- 3.4.3 The Bidder shall provide barbed wire across the tops of gates, as well as continuously along the entire top of the fence.

4. FENCING DETAILS

- 4.1 Post Setting: The Bidder shall provide posts of sufficient length for depths indicated on the approved Bidder's drawings.

- 4.2 Post Tops and Extension Arms: Malleable iron or pressed steel, with post tops for square posts designated to exclude moisture from inside the posts, and with integral extension arms for barbed wire, as follows:

- 4.2.1 For Intermediate Posts: Extension arm or arms extending at 45⁰ angles and slotted to carry three evenly spaced strands of barbed wire with the top strand or strands located approximately 35 cm above the top of the fence panel.

- 4.2.2 For End and Corner Posts: Same construction as for intermediate posts, with the post top and extension securely riveted to posts, or extend posts up to the full height required and provide a post top.

- 4.2.3 Gate Post: Extend posts up to the full height required and provide post top.
- 4.3 Top Rail Details: Provide with 17.5 cm long outside sleeve type couplings every 6 m on center. The top rail shall pass through the base of the intermediate post tops and form a continuous brace from end to end of each stretch of fence. Securely fasten the top rail to terminal posts with heavy pressed steel connections.
- 4.4 Brace Details: Provide at both sides of corner posts, and between terminal posts or gate posts and first adjacent intermediate posts. The compression brace shall be pipe, same as the top rail, installed midway between the top rail and ground and extending from the braced post back to the intermediate post. Securely fasten the compression brace to posts with heavy pressed steel connections. Securely fasten the truss from the intermediate post back to the braced post with a 10 mm diameter truss rod complete with tightening nut or turnbuckle.

5. GATES

- 5.1 Gate shall be swing type at the switchyard and sliding type at the power plant entrance. The swing type shall be hinged to swing 180 degrees from closed to open and sliding type shall be rolled on rail, complete with frames, latches, stops, keepers, hinges, fabric and braces rail. Gate leaves shall have intermediate members-and diagonal truss-rods as required for rigid construction and shall be free from sag or twist. Joints between frame members shall be made by welding or by means of heavy fittings, and shall be rigid and watertight. Gate fabric shall be the same as fence fabric and shall be attached to the frame ends by stretcher bars, bolt hooks or other mechanical means.
- 5.2 Hinges shall be of a heavy pattern with large bearing surfaces and shall not be twisted or turned under the action of the gate. Latches shall be the plunger bar type, full gate height, and arranged to engage the gate stop, except that single gates less than 3 m wide may be provided with a forked latch. Latches shall be arranged for padlocking with a padlock accessible from both sides of the gate. Stops shall consist of a roadway plate with an anchor set in concrete and arranged to engage the plunger.

- 5.3 Keepers shall consist of mechanical devices for securing and supporting the free end of the gates when in the full open position. Keepers shall be mounted on 70 mm outside diameter pipe posts filled with concrete and set in concrete foundations.
- 5.4 Gates shall be installed so that they cannot be removed without disassembly of the hardware. Hardware attachment bolts shall be pinned so that removal will be difficult.

6. FENCE CONSTRUCTION

- 6.1 The installed fence shall conform to the alignment and finish grade indicated on the approved drawings. Unless otherwise indicated or specified, all posts shall be plumb and spaced 2.40 m apart. Where necessary, the fence grade shall be adjusted to fit the ground contour by slipping the fence fabric links. On step grades, the posts may be set normal to the slope, provided transition sections are constructed. Ground surface irregularities shall be machine graded as required to eliminate frequent change in vertical alignment and to provide a smooth profile for the fence. Clearance below the bottom of the fence shall not exceed 100 mm.
- 6.2 It is the intent of these specifications that the plan and profile of each reach of fence between corner posts, or between corner posts and gate posts, shall be straight. The Bidder shall grade ground surfaces as required to achieve the straight profiles before each reach of fence is installed.
- 6.3 Where posts are set in earth, concrete foundations 90 cm deep shall be provided. Concrete foundations shall be not less than 35 x 35 cm for line posts and with a diameter not less than the post size plus 20 cm for terminal and gate posts. Foundations shall extend above the ground surface and shall be crowned approximately 50 mm. Concrete for foundations shall conform to the cast in place concrete section. Each foundation shall be cured for at least 72 hours before further works is done on the post.

7. GUARD RAIL

7.1 Materials

7.1.1 The rail elements shall be corrugated sheet steel beams as shown the Bid Drawing. All guard rail elements, terminal sections and fittings shall be interchangeable with similar parts as shown the Bid Drawing, regardless of source or manufacturer. The guard rail, terminal section and appurtenance shall be in accordance with AASHTO Designation: M 185, the plans and the following requirements:

- Unless otherwise specified, the beam shall be Class B, with buffer end sections being either Class A or B.
- The beams shall be galvanized after fabrication with type 1 or 2 coating in accordance with AASHTO Designation: M-180. The fabrication shall file a Brand Registration and Guarantee in accordance with AASHTO designation: M-180. Certificates of analysis shall be furnished for each per shipment covering all tests and measurements required by AASHTO Designation: M-180.

7.1.2 Guard Rail Posts and Spacer Blocks

- 1) Railing posts shall be of treated timber. Spacer blocks shall be of the same material and with the same treatment as the post. Timber posts shall be treated timber of Damar Laut and shall be of the section and length as specification or as shown on the approved drawings
- 2) The timber is treated by Creosote, Creosote Coal-Tar Solution or Pentachlorophenol Petroleum Solution. Treatment shall be 251 kg per cubic meter of timber and the type preservation to be used is as follows:
 - Creosote Oil – AWWA P1 for regular treatment; AWWA P13 for marine treatment
 - Creosote Coal – Tar Solution – AWWA P2, Grade A, for regular treatment; AWWA P12 for marine treatment. The ratio shall be 80% Creosote Oil and 20% coal-tar.
 - Pentachlorophenol – Petroleum Solution – AWWA P8 and P9.

7.2 Construction

The installed guard rail shall conform to the alignment and finish grade as indicated on the approved drawings. Unless otherwise indicated or specified, all guard rail posts shall be plumb and spaced 4 meters apart. Where posts are set in earth, concrete foundations shall be provided. Concrete for foundations shall conform to the requirements specified in Specification S-340 "Concrete Works". Concrete shall be GRADE 25. The length, depth and other sizes of the guard rail, guard rail post and spacer blocks shall be as indicated on the Bid Drawing.

8. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

- 8.1. Quality control shall conform to the requirements of Part 4, Subsection 2.7.
- 8.2. The quality control procedures and documentation.

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